MIL-D-36449A
15 December 1978
SUPERSEDING
MIL-D-36449
25 August 1971

MILITARY SPECIFICATION

DRIER, X-RAY FILM, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of top loading, portable drier for medical and dental x-ray film (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/6	- Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024
QQ-A-225/8	- Aluminum Alloy Bar, rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/8	- Aluminum Alloy, 5052 Plate and Sheet
TT-E-529	- Enamel, Alkyd, Semi-gloss.
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated, Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6525

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MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-L-10547	- Liner, Case and Sheet Overwrap, Water Vaporproof, or Waterproof, Flexible
MIL-M-7298	- Manual, Technical, Commercial Equipment

STANDARDS

FEDERAL

FED-TEST METHOD-STD-151	- Metals, Test Methods.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

Drawing No. 10183 Drier, X-Ray Film

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 73 Motor-Operated Appliances
UL 544 Medical and Dental Equipment

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, LI, NY 11749, 207 East Ohio Street, Chicago, IL 60611, 1655 Scott Blvd., Santa Clara, CA 95050 and 333 Pfingsten Road, Northbrook, IL 60062.)

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3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Fire and casualty hazards.

3.2.1 Proof of compliance. The contractor shall provide proof, to the procurement contracting officer, not less than thirty days prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standard of Safety 73. A copy of the test report shall be made available to the Government inspection personnel for their use in determining that the units being submitted for acceptance are identical to the unit submitted for approval. Proof shall consist of one of the following:

A. UL listing and UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, and the Government inspection personnel, upon request.

B. A label or equivalent means of identification affixed to the product as evidence that the product design has been examined and tested for design safety compliance by a nationally recognized independent laboratory. If an independent laboatory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. In support of the label/product identification, the independent testing laboratory shall provide surveillance over the product manufacturing system to the degree necessary to insure that production units are manufactured in the same manner and with the same type of components as the test sample evaluated and upon which the validity of the product identification is based. The testing laboratory shall develop a detailed test report which includes a conclusion as to whether the test sample evaluated is "approved" or "not approved" as complying with all requirements of the standards of safety.

3.3 Materials. Materials shall be as specified in the Bill of Materials of DPSC Drawing No. 10183. Where commercial components are specified, it shall be inerpreted to mean that similar components produced by other manufacturers may be substituted, provided that the component furnished is interchangeable with the specified component and that all the established functional requirements shall be met Prior to use of components other than those specified in the Bill of Materials, approval shall be obtained from the contracting officer.

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3.3.1 Corrosion-resisting steel. Corrosion-resisting steel shall comply with AISI 300 series of FED-STD-66 when tested as specified in 4.4.1.

3.4 Construction. The drier shall be of flush-type construction, for protection against mechanical damage or bodily injury. The overall dimensions shall be as shown on the drawing. The unit weight shall not exceed 45 pounds. The top, sides and door of the unit shall be insulated. The drier shall be of the top loading type having a positive acting door latch. Shall accomodate eight 14 by 17 inch radiographic films in standard size x-ray film processing hangers or twelve dental x-ray film processing hangers (see 6.1). Shall operate on 115 volt, 50-60 Hertz (Hz), AC. The threads of all screws and bolts shall be coated with a locking compound to prevent them from becoming loose during usage. Shall be equipped with a replaceable air filter. Construction shall be in accordance with DPSC drawing 10183.

3.5 Final assembly. After finishing as specified in 3.8, thermal insulation shall be secured in place by adhesive or double-faced tape as shown on the drawing. Electrical components shall then be installed, and final assembly of the unit completed. Where grounding is required, electrical parts shall be grounded to clean, bare metal.

3.6 Performance. Drying shall be accomplished by means of forced circulation of heated air past the film. The drier shall be of the recirculating type, with air temperature thermostatically controlled. Thermostat shall be set to provide a temperature of $140^{\circ} \pm 5^{\circ}$ Fahrenheit (F.) when tested as specified in 4.4.2. The unit shall have sufficient capacity to dry a full load of eight 14 inch by 17 inch radiographic films in 33 minutes or less when tested as specified in 4.4.3.

3.6.1 Dielectric withstand. There shall be no evidence of electrical breakdown when tested as specified in 4.4.4.

3.7 Color. The color of the x-ray film drier shall be No. 24087 of FED-STD-595 (see 6.3).

3.8 Finish.

3.8.1 Primer. Prior to final assembly of the unit and installation of electrical components and insulation, the surfaces of all aluminum-alloy subassemblies shall be thoroughly cleaned and chemically degreased, and then treated in accordance with class 2 of MIL-C-5541.

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3.8.2 Enamel. All metal surfaces, except corrosion resisting steel, plated hardware and those to which insulation is to be applied shall be coated with semi-gloss enamel conforming to class A of TT-E-529 to a total dry film thickness of 1.6 to 2.0 mil.

3.9 Identification marking. Each drier shall have a metal nameplate permanently marked with the following information, and located to permit reading without difficulty:

Manufacturer's name or registered trademark.
Manufacturer's model or serial number.
Contract or purchase order number.
National Stock Number (NSN).
Electrical rating

As an alternate, marking may be stamped into the metal in an area which can be observed by the user.

3.10 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies shall be at the option of the Government.

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3.11 Welding. Welding shall be durable, sound in every detail and in accordance with the latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Welds shall be smooth and free of slag, excessive undercut, or excessive splatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds, except spot welds, shall be continuous with no visible voids, pin holes, or cracks

3.12 Workmanship. Workmanship shall be first class throughout. The x-ray film drier shall be free from defects which detract from its appearance or impair its serviceability. The unit shall be free of sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on one unit when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component/materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government, and provided also that the test methods and the results obtained comply with those described in this specification. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method. individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one drier.

TABLE I. Sampling for examination.

	Inspection level	AQL (defect per hundred units)
For visual examination		
Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0
For dimensional defects	S-4	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for test purposes shall be one drier. Where practicable, one or more test characteristics shall be conducted on the same sample unit.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection
Cut-off Temperature	3.6	4.4.2	100%
Performance	3.6	4.4.3	S-1
Dielectric withstand	3.6.1	4.4.4	100%

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the manufacture of the drier shall be selected for chemical analysis.

4.3 Examination. The drier shall be examined to determine compliance with all requirements contained in this specification

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Not complete with all required parts, subassemblies, and materials.
102	Components and parts not free of cracks, tears, ruptures, and deformed surfaces.
103	Components, mating parts, loose or not properly aligned and fitted.
104	Seam or fillet welds; burn holes, cracked, fractured or otherwise not fused.
105	Spot welds not sound.
106	Exposed or accessible edges, corners, and surfaces not free of burs, sharpness, and sharp projections.
107	Solder or brazing not adherent.
108	Components in the electrical system not in accordance with requirements.
109	Electrical systems in units not consistent in layout and neatly wired.
110	Insulation of wire connectors not clear of, or protected from, framework edges to avoid occurrence of shock hazard or shorts due to vibration.
111	Electrical wiring not free of cracks, breaks, and cuts in the insulation.
112	Electrical system not free of taut leads, loose or improperly fastened terminal connections, and poor solder joints.
113	Ground wire not connected to clean, unpainted metal.
Minor	
201	Fasteners not tight.
202	Finish not free of chipped areas, cracks, scratches or roughness.
203	Marking not present, properly located, legible, correct or permanently affixed.
204	Service data not present in duplicate.
205	Unit not clean.
206	Soldering or brazing not clean.
207	Paint not free of sags, runs and thin areas.

4.3.2 Dimensional examination. The drier shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

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4.3.3 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. During tests specified in 4.4.2 and 4.4.3, the input voltage shall be 115-120 volts, 60 Hz, AC.

4.4.1 Chemical composition. Four ounces from each lot of material shall be tested in accordance with FED-TEST METHOD STD-151.

4.4.2 Cut-off temperature. The sensing element of a temperature measuring instrument shall be introduced into the empty drying compartment of the unit, and secured at the geometric center. The drier shall be operated until the thermostatically controlled heater is cycling, and for at least 20 minutes thereafter. During this period, temperature readings shall be taken each minute. (If a recording-tape instrument is used, the recorded chart is acceptable in place of the temperature readings.) Data obtained shall be plotted and analyzed to determine whether temperature falls within the tolerance specified in 3.6 Unit shall be examined to insure that all components are operational during this test.

4.4.3 Performance. A full load of eight 14 by 17 inch x-ray films shall be exposed to white light and developed. The processed wet films shall be inserted into standard film hangers and shaken to remove excess water accumulation. The hangers shall be loaded into the drier which has been preheated to the extent that the thermostatically controlled heater is cycling. Average ambient conditions for the duration of the test shall be maintained at 68° - 72°F. and 50-55 percent relative humidity. At the conclusion of the test, the films shall be examined for water residue. Water droplet accumulations resulting from hanger run-off shall not be considered a failure.

4.4.4 Dielectric withstand. A dielectric withstand test shall be conducted in accordance with requirements of UL 544. The test sample shall be observed for evidence of electrical breakdown.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6 2).

5.1.1 Unit of issue. One drier, as specified, constitutes one unit of issue.

5.1 2 Level A. Each unit shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. The line cord shall be neatly coiled and attached to the unit in such a manner as to prevent unraveling. Closure shall be as specified in the applicable box specification.

5.1.3 Level C. Each drier shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified, (see 6 2).

5.2.1 Level A.

5.2 1.1 Shipping container. Each unit shall be overboxed in a shipping container designed for a type 1 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601. overseas type; or PPP-B 621. class. Closure and strapping shall be as specified in the appendix of the box specification.

5.2 1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.2 Level B.

5.2 2.1 Shipping container. Each unit shall be overboxed in a shipping container designed for a type 1 load and constructed in accordance with PPP-B-585, class 3. style 3; PPP-B-601, overseas type; or PPP-B-621, class. Closure and strapping shall be as specified in the appendix of the box specification.

5.2.3 Level C. Each packaged drier shall be packed in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

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5.3 Marking. Marking shall be marked as specified in MIL-STD 129.

6. NOTES

6.1 Intended use. The portable x-ray film drier shall be suitable for use in ships, hospitals and field medical units. Shall accomodate up to eight 14 x 17 inch medical film hangers, or twelve NSN 6525-00-603-1795 dental film hangers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information on the procurement documents:

a. Title. number and date of this specification.
b. National Stock Number.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When a first article is required (see 3.1, 4.1.2.1 and 6 4).

6.3 Availabilty of color chips. Color chips are available by referencing the color number and FED-STD-595 from

Commanding Officer
U.S. Naval Publicetions & Forms Center
5801 Tabor Road
Philadelphia. PA 19120

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Defense Acquisition Regulation. The contract or purchase order specifies the quantity. procedure for delivery and other specific instructions for inspection and approval of the first article.

6.5 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6525-00-664-0858	DRIER, X-RAY FILM, Portable. Lightweight. 115 volt. 50-60 Hz. AC

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6 5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA - DM

Project No 6525-1134

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-36449A DRIER, X-RAY FILM, PORTABLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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